A Literacy-Free, Report-Free Eye Exam on a Consumer EEG Headband and VR Headset: Toward Objective Vision Screening by Community Health Workers

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***Abstract*—At least a billion people have a preventable or unaddressed vision impairment, and much of the problem is screening: a letter chart needs a subject who can read it and an examiner who can judge the answers. We built an eye exam in which the subject only looks at a dot with one eye covered. It runs on a Muse headband with one extra electrode over the visual cortex and a Meta Quest 3S that shows the stimulus in its web browser, for under 600 US dollars in total. Twelve tests are read from the steady-state visual evoked potential (SSVEP): acuity, vernier acuity, contrast and its sensitivity function, visual field, gaze, four meridians for astigmatism, red/green flicker photometry, colour, the light threshold, and the eye's temporal tuning and apparent latency. A synthetic subject with known deficits passes all 39 checks of an end-to-end test of the software, so the system can be checked without a patient. On one person, the first three sittings found nothing because the headset's browser was running an old cached copy of the page. Once we verified what the headset actually displayed, the next three sittings gave a flash response in both eyes, an acuity limb down to 6 arcmin that repeated in the same eye six days later, a response at every gaze position, and a red/green photometric null at the same mix in both eyes. They also showed that a loose electrode can pass a significance test without carrying the visual response. On a laptop panel at a measured distance, the reversing bars gave a grating acuity of about 20/18 and 20/25 in physical units, finer than the headset can draw, and a light ladder put both eyes' thresholds at 0.5% of white. We also describe a hearing test on the same headband, which tags four octave bands of sound at rates near 40 Hz and reads each band's auditory steady-state response behind the ears. The results print in 3D as a raised letter chart and a tactile report, and photometry with a slow red/green swap could screen colour vision objectively. The system is a research prototype, not a diagnostic device.**

***Index Terms*—mobile health; visual screening; hearing screening; SSVEP; ASSR; sweep VEP; wearable sensors; consumer electronics; community health workers.**

I. Introduction

The WHO counts at least 2.2 billion people with a vision impairment, at least a billion of whom have a condition that could have been prevented or has not been addressed, with uncorrected refractive error the leading cause [1]. The lenses are cheap. The screening is the bottleneck. A letter chart only works if the subject can read the examiner's alphabet and the examiner can judge the answers. Other screens have reached community health workers by pairing a consumer device with a trained worker and software that checks itself [2], [3]. We wanted the same for vision.

The steady-state visual evoked potential (SSVEP) gives an objective way in. A pattern that flickers at a fixed rate drives the visual cortex at that rate, and the response can be read from the scalp without the subject saying anything. The sweep VEP was built on this idea to measure acuity in infants, who cannot report [4], and the same approach gives a contrast threshold [5]. An exam read from the brain removes literacy, language and examiner judgement from the test, although it brings its own biases, such as hair and headband fit [6].

This paper makes four contributions. We describe a twelve-test eye exam that runs on a consumer EEG headband and a consumer VR headset (Section II). We validate the whole software stack with a synthetic subject, so the device can be tested with no patient in front of it (Section III). We report six sittings on a person, including the failures and what they changed (Section IV). Finally, we describe a hearing test on the same headband (Section V).

II. The Rig and the Exam

Mann et al. first used the SSVEP to turn the eye into a camera [7]. We rebuilt that rig from consumer parts (Fig. 2). A Muse headband records four dry electrodes plus a gold cup electrode that we added at Oz, over the visual cortex. A phone streams the raw EEG to a laptop, acting as the hub in the wireless sensing architecture that iCalm set out [8]. A Meta Quest 3S shows the stimulus as a web page in its browser, and the laptop drives the page, logs every frame and scores each test (Fig. 3).

Every stimulus fills the whole window (Fig. 1). Our earlier camera experiments had shown that the response grows with the flickering area and that a small square gives almost nothing. Each eye starts with the window flickering between white and black, the same stimulus the camera used for calibration, in five ON/OFF pairs of 4 s at 12 Hz on the 72 Hz display. This flash has to produce a response before anything else is read, and its best channel and amplitude become the reference for the rest of that eye's tests. The flash then runs at six rates from

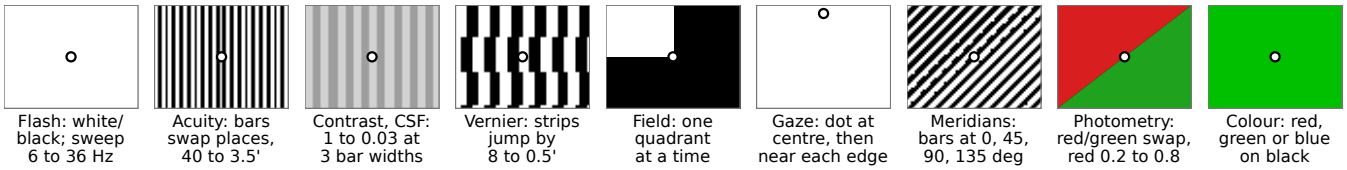


Fig. 1. What the headset shows during each test, drawn as the page draws it. Every stimulus fills the browser window and the subject keeps their eye on the dot. ON blocks of 4 s, with the window flickering or the bars swapping places at 12 Hz, alternate with 4 s OFF blocks (grey for the gratings, black otherwise). The vernier offset is exaggerated here so it shows in print.



Fig. 2. The rig: the Quest 3S worn over the Muse headband, with the cup electrode held at Oz by the headset strap.

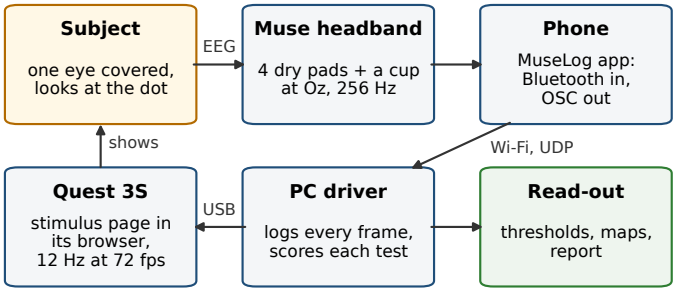


Fig. 3. How the exam runs. The headband's EEG reaches the laptop through the phone, the laptop drives the page in the headset over USB, and the subject looks at the page. In the software test of Section III, a synthetic EEG that follows the logged stimulus replaces the subject and the headband.

6 to 36 Hz. The amplitude across rates gives the eye's temporal tuning, and the slope of the response phase across rates gives its apparent latency [9]. The difference in latency between the two eyes is the main measure of the clinical pattern VEP.

The other tests use two ON/OFF pairs per step. For acuity, a black and white grating has its bars swap places at the flicker rate, so each resolved bar flickers like the flash while the mean luminance stays constant (pattern reversal [4]). The bar width shrinks from 40 to 3.5 arcmin. The contrast test uses one bar width and lowers the contrast from 1 to 0.03. The field test flickers one quadrant at a time. The gaze test flickers the whole window again while the dot sits at the centre and then 84% of the way to each edge, so the flicker falls to one side of the fovea. The colour test flickers the window red, green or blue against black.

We added four tests after the fourth sitting. The vernier test cuts coarse bars into strips three bars high, and every other strip jumps sideways by an offset and back at the flicker rate. The offset shrinks from 8 to 0.5 arcmin, and the canvas's anti-aliasing delivers it to a quarter of a device pixel. This is the vernier VEP, a hyperacuity that amblyopia degrades more than ordinary acuity [10]. The contrast sweep is then repeated at a coarse and a fine bar width, which gives three points on the contrast sensitivity function (CSF). The meridian test turns a 14 arcmin grating to 0, 45, 90 and 135 degrees, because a weak meridian next to a strong one at right angles is the signature of astigmatism. The last test is heterochromatic flicker photometry [11]. A red field and a green field swap at the flicker rate while red's share of the pair's luminance runs from 0.2 to 0.8, over a steady mixture of the two as the OFF screen. The response vanishes where the fields look equally bright to this eye; that red share is its isoluminant point (0.5 for the display's standard observer, higher for a red-weak eye), fitted as $a\,M(s;p_0)+b$ rather than read off the dimmest step.

The sixth sitting added a light ladder: the flash at eight luminance levels from full white down to 0.8% of it, in linear light, whose dimmest followed level is the eye's light threshold. The headset's LCD is never fully dark, so this measures daylight sensitivity, not night vision. The exam also runs on the laptop's own 2560-pixel, 240 Hz panel at a measured distance, where the pixels per degree follow from the panel's width: at 1.5 m one arcmin is 3.2 device pixels, while the headset, at about 20 pixels per degree, gives each arcmin a third of a pixel.

Each step's ON blocks are scored against all of the test's OFF blocks by the line power at the flicker frequency and its harmonic, with an exact permutation test over the channels in contact. Contact has two tiers. A channel above 300 μV rms is open or railing and is set aside. A channel between 60 and 300 μV is noisy and is kept only if its own ON/OFF test is significant. Thresholds follow the sweep-VEP rule: a line is fitted through the falling limb, from the step just below the finest one that responds up the rise, and extrapolated to zero amplitude. Two requested bar widths that the display delivers as the same width count as one step.

We also check that the stimulus was really shown. The page is served with caching disabled at a fresh URL for each sitting, its code is fingerprinted through the browser's debug port before the subject starts, and the headset's own frame buffer is captured during a block. With all twelve tests the exam takes about seventeen minutes per eye, and the driver can run any subset.

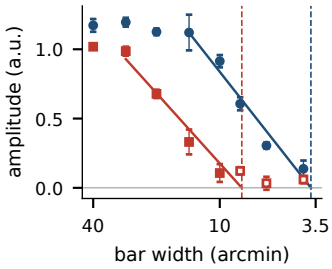


Fig. 4. Acuity on the synthetic subject (open: $p \geq 0.05$), limbs and zero crossings (dashed): 3.7 and 7.9 arcmin.

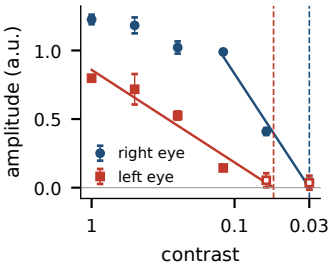


Fig. 5. Contrast on the synthetic subject at one bar width: 0.030 and 0.053.

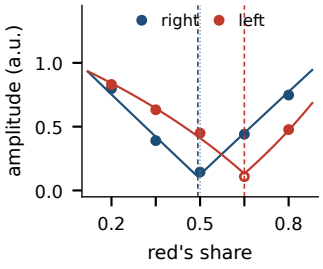


Fig. 6. Flicker photometry on the synthetic subject: fitted nulls (dashed) at 0.49 and 0.65; the display's match dotted.

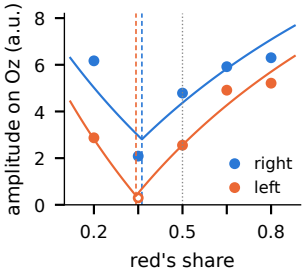


Fig. 7. The same on a person (sixth sitting, Oz; open: not significant): nulls at 0.34 (left) and 0.36 (right).

III. Testing Without a Patient

A device meant for a worker with no eye clinic nearby has to be testable without a patient. We wrote a synthetic subject that reads the stimulus log as it is written and streams EEG whose SSVEP follows what is on the screen, with per-eye cut-offs. The gate's right eye stops resolving bars at 8 arcmin, contrast at 0.08 and vernier offsets at 1 arcmin, has one dead quadrant, responds 120 ms late, holds up to 40 Hz and has a weak horizontal meridian (0.4 of full, rising as $\cos^2$). The left eye is twice as coarse, whole, 150 ms late, half gone by 24 Hz and red-weak (its fields match at a red share of 0.65, with a tenth of the response left at the null); the colour gains are 0.7, 1 and 0.4. The gate runs the full exam on the laptop's 60 Hz panel at 15 Hz and asks 39 questions, from whether every stage was delivered to whether each threshold falls inside the band where the synthetic eye's visibility runs from 0.02 to 0.6.

All 39 checks pass. The right eye's acuity threshold is 3.7 arcmin and the left eye's 7.9, more than twice as coarse (Fig. 4). The contrast thresholds are 0.030 and 0.053 (Fig. 5). In the right eye, exactly the dialled quadrant is silent. The apparent latencies are 109 and 139 ms, within 6 ms of what the gate expects for the dialled delays and 30 ms apart as dialled. The right eye is still above half amplitude at the fastest rate, and the left falls to half at 22.5 Hz. Every gaze position responds at 0.92 to 0.99 of the centre in both eyes, and green beats red, which beats blue.

The four newer tests pass their checks too. The vernier thresholds are 0.37 arcmin for the right eye and 0.88 for the left. The right eye's response falls from full at 2 arcmin, through half at 1, to nothing at 0.5, exactly the dialled curve, which also shows that the page really delivers sub-pixel offsets. Every CSF threshold falls inside the contrast band. The right eye's weakest meridian is at 90 degrees, at 0.47 of the best, while the left eye's four meridians stay within 0.90 of each other. The photometric nulls land at 0.49 and 0.65, a difference of $+0.16$ (Fig. 6). What the gate cannot judge is a stimulus's strength. A letter chart of Landolt Cs passed it and then gave nothing on a person (Section IV): the synthetic subject responds to what is visible, not to how much of the screen changes.

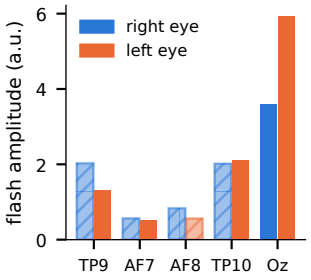


Fig. 8. Fourth sitting: the flash on each channel (hatched: not significant). Oz responds in both eyes.



Fig. 9. Fourth sitting: each quadrant over the best. Both eyes show the same pattern.

IV. Six Sittings on a Person

One of the authors ran the exam four times in one day, right eye then left, at 12 Hz, a fifth time the next day, and a sixth five days after that. The first three sittings fitted nothing, and the first taught us to check contact. Both ear electrodes had sat at 100 to 280 μV rms, loose under the headset strap, and the signal check, which only looked for electrodes stuck at the amplifier's rails, had passed them. Such quiet failures are what a consumer wearable brings to the field [3]. The exam now measures every channel before each eye and tells the subject which pad to press.

The cause of all three null sittings showed up in the fourth, in the headset's own frame buffer. During an ON block the view was black except for the dot, and the page's code, read through the browser's debug port, did not contain the flash test. The browser had kept a cached copy of the page from an earlier build, while the driver logged 72 frames per second of flicker from the page it thought was running. A logged stimulus is not necessarily a shown one, which is why the exam now checks what the headset displays (Section II).

With the right page loaded, the fourth sitting gave a flash response on the occipital electrode in both eyes: $p = 0.024$ at a line SNR of 9.6 in the right eye and $p = 0.008$ at 18 in the left (Fig. 8). The right eye responded best at 7.2 Hz (half gone by 8) and the left at 12 Hz (holding to 25). The left eye resolved bars down to 6 arcmin, with the limb extrapolating to 2.5 arcmin, while the right eye only responded to 21 arcmin and coarser. Neither eye gave a contrast limb. Both eyes showed the same quadrant pattern, bottom-left strongest and bottom-right weakest or silent (Fig. 9). Two

retinas would not agree like this, but one electrode sitting off the midline between the two calcarine banks would. The colour ratios agreed between the eyes (R:G:B of 0.69:1:0.18 and 0.72:1:0.26), mostly reflecting the display's primaries.

This sitting changed two parts of the method. The left eye's occipital electrode sat at 105 μV rms, which the old 60 μV rule set aside, yet it carried the sitting's strongest line (SNR 6 to 18 in every ON block, under 1.5 in every OFF block): hence the two contact tiers. And the apparent latencies, 77 and 258 ms, differ by a full turn of phase, not by physiology, because the fit chose different turn counts for eyes that responded at different rates; the latency stays undetermined until one turn count is shared.

The fifth sitting ran all eleven tests on the right eye. Its flash gave 3.0 ($p = 0.004$). The sweep peaked at 18 Hz and held to 27, where the day before the same eye had peaked at 7.2 and was half gone by 8, so the right eye's temporal tuning is not yet stable. The bars responded down to 6 arcmin (limb at 2.7) against 21 the day before, which we put down to an electrode that stayed in contact, and every gaze position responded, at 0.3 to 0.9 of the centre. The vernier, contrast and CSF tests gave no limb, with the occipital cup in the noisy tier. Twice the phone's stream stopped, and the driver carried on through five tests with no EEG; we reran them and joined the two sessions.

The sixth sitting completed the left eye and began with a failure that the contact tiers let through. With both ear electrodes near 265 μV rms and Oz at 186, all in the noisy tier, the flash gave nothing on Oz ($p = 0.57$) but a line on TP9 (2.0, $p = 0.004$), which became the reference. Oz then came off entirely. Reseated (23 μV), it gave a flash of 4.67 at a line SNR of 77, and TP9 gave nothing ($p = 0.39$), here and in the eye's later run. A line on a loose electrode that vanishes once Oz is in contact is most likely the display's flicker picked up through the strap, so the reference should come from Oz unless it is open.

With Oz in contact, the left eye repeated the fourth sitting's acuity limb (finest bar 6 arcmin, crossing at 2.5) and, in a separate sweep, its best rate (12 Hz, half by 23.6): the first measures to repeat across sittings. Three quadrants and all five gaze positions responded. Vernier, contrast, CSF and meridians again gave no limb, now with Oz in contact, so these weaker stimuli may need more blocks per step. Its apparent latency, 88 ± 5 ms, still carries the turn-count ambiguity. Photometry gave the first null on a person, at red shares of 0.34 and 0.36 in the left and right eyes, with 7% and 40% of the response left (Fig. 7). The two match within the 0.15 step, so the shift from the display's 0.5 most likely comes from its red primary, not the eyes. The plain colour ratios moved (now 0.76:1:0.50 right, 0.96:1:0.23 left), so photometry, which cancels luminance by design, is the better colour measure.

Later that day we tried a letter chart of Landolt Cs turning 180° at the flicker rate, so that only an eye resolving the gap would follow. No line responded, not even at 20/400, while the flash gave 4.4 to 5.8 on Oz: turning a C changes only its gap areas, about 2% of the screen, where reversing bars change

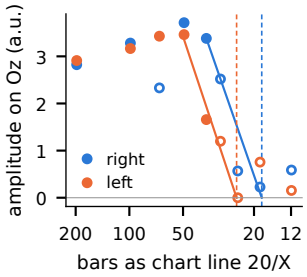


Fig. 10. Sixth day, laptop at 1.5 m: reversing bars at chart sizes, with limbs crossing at 0.9 and 1.3 arcmin (20/18, 20/25).

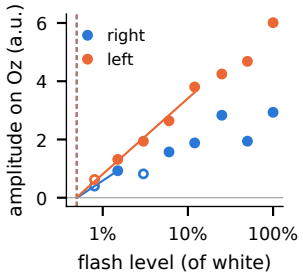


Fig. 11. Sixth day, headset: the light ladder, both eyes falling to a threshold near 0.5% of white.

all of it. The chart that worked was the bars themselves at the chart's sizes, on the laptop panel at a measured 1.5 m, where the thresholds are in physical units (Fig. 10). Both eyes responded strongly down to bars of 20/40 size, and the limbs cross at 0.9 arcmin in the right eye and 1.3 in the left, a grating acuity of about 20/18 and 20/25 that has not yet been calibrated against a letter chart. The headset could not have shown this: it draws nothing finer than about 20/60, so its 6 arcmin limbs sit near its own pixel. With four repeats per level (two, with Oz loose, could not be read), both eyes' responses fell steadily with the light, to thresholds of 0.50% and 0.49% of white (Fig. 11). Across the sittings the occipital contact, now read live while the cup is reseated, has limited the exam more than any stimulus.

V. A Hearing Test on the Same Headband

The same headband can test hearing. An auditory steady-state response (ASSR) is a brain potential that follows a periodic sound. When a sound's amplitude is modulated at a fixed rate f_m, the EEG gains a component at exactly f_m, which shows up as a narrow peak in the spectrum and can be detected objectively [12]. In awake subjects the response is largest near 40 Hz [13]. Its amplitude grows with modulation depth and loudness, so a threshold can be found by lowering the level until the peak disappears into the background [12].

We split a broadband sound into four octave bands centred at 500, 1000, 2000 and 4000 Hz and modulate each band at its own rate near 40 Hz: 37, 39, 41 and 43 Hz. Each band leaves its own peak in the EEG, so all four can be read at once from TP9 and TP10, the Muse electrodes just behind the ears [14]. Clinical ASSR audiometry usually uses 80 to 90 Hz, whose mostly subcortical generators keep working during sleep, which suits infants. In awake adults, responses near 40 Hz are two to three times larger and come mainly from the auditory cortex [12], [15]. That extra amplitude matters for dry electrodes, which are noisier than clinical gel systems, and the price is that the listener has to stay awake and alert. Clinical systems already test 500 to 4000 Hz carriers at the same time [16], and octave-wide bands have been used in place of pure tones [17]. As far as we know, nobody has yet obtained a frequency-specific EEG audiogram from a consumer dry-electrode headband, or used frequency-tagged music as the stimulus.

A. Recording

The Muse records at 256 Hz from four dry electrodes near TP9, AF7, AF8 and TP10, referenced to Fpz [18], and we analyse only TP9 and TP10, the closest to the auditory cortex. The headband streams over Bluetooth with `muselsl` into the Lab Streaming Layer, which also carries the stimulus events, so the EEG and the audio share one clock. Timestamps come from the headband's packet counter and are not dejittered, so a lost packet stays a gap. Ten seconds of resting EEG check contact first. The listener sits upright with eyes open on a fixation point, jaw relaxed, and is asked to stay alert, because 40 Hz responses weaken with drowsiness [12].

B. Stimulus

The source is a music track converted to mono at 44.1 kHz or, as a control, pink noise, whose energy is spread across frequency much like music. Fourth-order zero-phase Butterworth filters cut it into octave bands from $f_c/\sqrt{2}$ to $f_c\sqrt{2}$, and each band $x_b(t)$ is modulated at its tag rate:

$$s_b(t) = x_b(t)\left[1 + m\sin\!\left(2\pi f_m t - \tfrac{\pi}{2}\right)\right], \qquad (1)$$

with full depth ($m = 1$). The phase offset starts every envelope at zero, so each block begins in silence. Each band is scaled to an RMS of $10^{(L-100)/20}/\sqrt{2}$ of full scale for a target level L, where 100 dB is the level we assign to a full-scale sine, so all four bands play at the same level. The bands are summed with 20 ms ramps, capped at 85 dB per band, at a fixed system volume. The levels are not acoustically calibrated, so every level here is nominal and absolute thresholds do not transfer between devices. Comparisons between EEG and behavioural thresholds, measured on the same headphones and volume, are not affected. Attenuating one band by a known amount before mixing simulates a hearing loss with a known answer: a band cut by 20 dB should have its EEG threshold rise by 20 dB.

C. Protocol

A session has three stages. First, a go/no-go check plays a 1 kHz tone modulated at 40 Hz in 120 s blocks at 60, 50, 40, 30 and 20 dB. The session continues only if the two loudest levels give significant responses; otherwise we fix the contact, for example by moistening the ear pads, and repeat. Second, a behavioural audiogram at 500 to 4000 Hz uses a modified Hughson–Westlake procedure [19] on the same headphones: 1 s tones from 40 dB, down 10 dB after each tone heard and up 5 after each miss, with silent catch trials. Third, the four-band stimulus plays passively in 60 s blocks at 60, 50, 40, 30 and 20 dB, about five minutes in all.

D. Analysis

TP9 and TP10 are band-passed at 1 to 70 Hz, notch-filtered at 60 Hz and averaged. Each block is cut into 4 s epochs starting 2 s after onset, and epochs with a long dropout or over 150 μV peak to peak are dropped. The epochs are averaged coherently as complex spectra, so the phase-locked ASSR adds up while the background averages out. At each tag frequency, an F-test compares the power at the tag with the mean power of the K neighbouring bins within ± 6 Hz, leaving out the other bands' tags [20]:

$$F = \frac{P(f_m)}{\frac{1}{K}\sum_{k=1}^{K} P(f_k)}, \qquad (2)$$

and a response counts as present at $p < 0.05$. A band's EEG threshold is the lowest level with a significant response, provided every louder level is also significant, so one false positive at a quiet level cannot set it. ASSR thresholds usually sit 10 to 20 dB above behavioural ones [21], [22], so we compare the two mainly by correlation, alongside the per-band difference and the mean absolute error. On simulated EEG with known thresholds the pipeline recovered them to within one 10 dB step, with a false-alarm rate of 4.7% at a nominal 5%. Recordings on people are the next step.

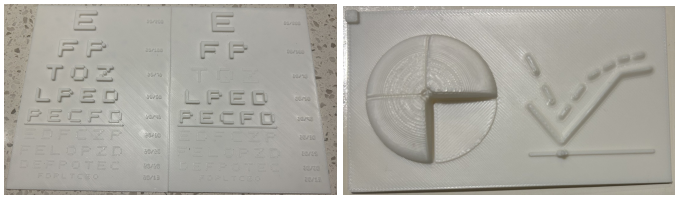


Fig. 12. The sixth day's results printed in PLA. Left: the letter chart as the cortex responded on the laptop at 1.5 m, left eye then right; each line's letters stand as high as the response at that line, and a bar marks the finest line that responded (20/40 in both). Right: a tactile report, the left eye's four quadrants as a dome (the low quadrant did not respond) and both eyes' red/green photometry as ridges, left solid and right dashed, with a peg under the match.

VI. What the Exam Gives, Printed

Readings. One sitting of about seventeen minutes per eye gives twelve thresholds and maps with no spoken answer. A letter chart gives one, acuity, only from a subject who reads the examiner's alphabet and will answer, and it reports the last line read; a young child, a reader of another script or a person with aphasia may give no usable answer. Here every step carries a significance test, both eyes share one scale, and a repeat sitting follows change. None of it is yet validated against clinical tests.

Colour without plates. Plate tests ask the subject to name a number hidden in coloured dots, so they need the digits, good printing and good light, and they give pass or fail. Flicker photometry asks nothing: the red share at which a red/green swap stops driving the cortex is the eye's own brightness match, and a red-weak eye needs more red (0.65 against 0.49 in the synthetic subject). At that match, a slow red/green swap still changes hue for a normal eye, which the cortex answers with a chromatic VEP [23], while an eye that cannot tell red from green sees nothing change and gives no response [24]. No response to a swap made isoluminant for that eye is thus an objective sign of red-green deficiency, graded by how far the match moved. On a person we have the match (0.34 and 0.36); the slow swap is next.

Printing the result. "1.3 arcmin" means little to most people, so we print the result on a consumer 3D printer (Fig. 12). On the familiar chart each line stands as high as

that eye's response to bars of its size, so a person sees which lines their brain followed and where it stopped; the tactile report raises the field as a dome, the silent quadrant low, and the colour match as a ridge with a peg at the null. A print can be held, kept and compared with the next screening's, and read by touch it reaches people who cannot read a chart, including those with low vision whom a screen exists to find. It also shows noise honestly: the right eye's 20/70 line stands low between lines that responded, a dropout a pass/fail chart would hide.

VII. What Is Established and What Comes Next

So far we have built and checked the apparatus: an exact test for every step, the sweep-VEP threshold rule, the latency from the phase slope, the photometric null, and a 39-check software gate that catches failures without a volunteer. On one person we have a flash response in both eyes, each eye's best flicker rate, an acuity limb down to 6 arcmin and a response at every gaze position, with the left eye's limb and best rate repeated six days apart, a photometric null at the same mix in both eyes, a light threshold of 0.5% of white in each eye, and, on the laptop, a grating acuity of about 20/18 and 20/25 in physical units. The same recording carries more than the visual response [25]: in companion sittings the forehead electrodes gave the blink rate, which doubled between two sittings in one evening, a long-known sign of fatigue [26] and a natural cue for a just-in-time prompt to rest [27].

Several things are not yet established. The headset's thresholds are not in physical units and its finest bars are display-limited; on the laptop they are, but grating acuity still has to be calibrated against a letter chart. The latency still has its turn-count ambiguity. We do not yet have a vernier or contrast reading on a person, even from a sitting with the occipital electrode in contact, and the photometric null is only relative until the display's primaries are measured. The exam does not yet pause when the stream drops, nor insist on the occipital electrode for the reference. The hearing test still has to be run on people. After that come a blur study with trial lenses against a letter chart, a normative group, and a flow for health workers with the report on one page or in print. Nothing here is a diagnosis yet, but every part of the exam can now be tested, by a person or by the synthetic subject, on hardware that costs under 600 US dollars.

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